Homework — 1.2.2 Applications Generation

Name: ______________________ Date: __________ Mark: ______ / 20

OCR H446 — set after the 1.2.2 lesson. Show your working.

Part A — This week's topic (~14 marks)

1. Match each term to its definition by writing the correct **letter** in the Match column. Two definitions are not needed. **[3]** *(AO1)*

Term	Match
Application software	
Utility software	
Open source software	
Closed source software	
Linker	
Loader	

Letter	Definition
A	Copies a program from secondary storage into RAM and prepares it to run
B	Software whose source code is freely available to view, modify and redistribute
C	Software that maintains or optimises the system, e.g. compression, backup, antivirus
D	Translates and executes a program one line at a time
E	Software that performs a task the end user needs doing, e.g. word processing
F	Combines compiled modules and library code into a single executable program
G	Software distributed as an executable only, under licence; the source code is kept secret
H	Small, very fast memory located on or near the CPU

2. The four stages of **compilation** are shown below in a jumbled order. Number them **1–4** to show the order in which they happen. **[2]** *(AO1)*

Stage	Order (1–4)
Code generation	
Lexical analysis	
Optimisation	
Syntax analysis	

3. The **code generation** row of the table has been completed as an example. Complete the description of what happens during each of the other three stages of compilation. **[3]** *(AO1)*

Stage	What happens
Code generation *(example)*	Object code (machine code) is produced from the analysed program
Lexical analysis	(a)
Syntax analysis	(b)
Optimisation	(c)

4. Each statement in the table below contains **one** factual error about linkers, loaders and libraries. Identify the error in each and rewrite the statement correctly on the lines below. **[3]** *(AO1)*

#	Statement (each contains one error)
1	When a compiled program uses code from a library, the loader combines the compiled modules and the library code into a single executable file.
2	With static linking, the library is kept as a separate file and connected to the program at run time, so updating that one file updates every program that uses it.
3	An advantage of dynamic linking is that the executable contains everything it needs, so it cannot fail because a library file is missing from the user's computer.

5. A school is choosing between **open source** and **closed source (proprietary)** office software. State **two** advantages of choosing open source and **one** advantage of choosing closed source. **[3]** *(AO2)*

Part B — Retrieval: keep it fresh (~6 marks)

6. Describe, in order, the steps the CPU and OS take to handle an **interrupt**, making clear why a **stack** is used. **[3]** *(1.2.1)*

7. Explain **one** way in which a **RISC** processor differs from a **CISC** processor. **[1]** *(1.1.2)*

8. State Von Neumann architecture's key principle that distinguishes it from Harvard architecture. **[2]** *(1.1.1)*